

Experiment 7: Depth First Search (DFS)

Aim

To implement the Depth First Search (DFS) algorithm for graph traversal.

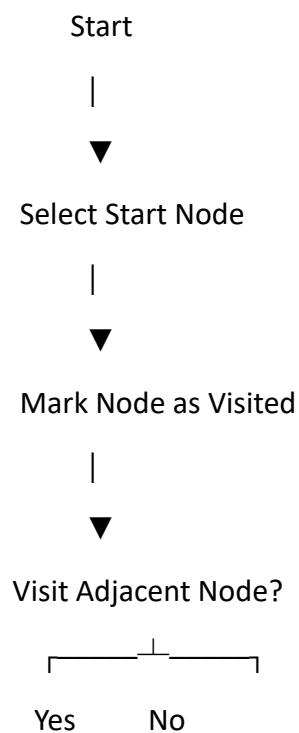
Objective

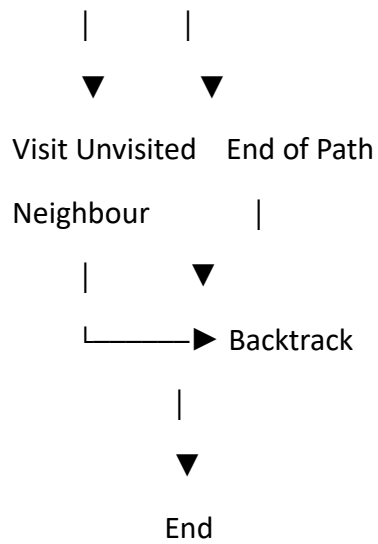
To traverse all the vertices of a graph using the Depth First Search technique.

Algorithm

1. Start from the source vertex.
2. Mark the current vertex as visited.
3. Visit the current vertex.
4. Recursively visit all adjacent unvisited vertices.
5. Continue until all vertices are visited.
6. Display the traversal order.

Flowchart





Python Code

```

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

    for neighbour in graph[node]:
        dfs(neighbour)
  
```

```
dfs('A')
```

Sample Output

```
A B D E F C
```

Result

Thus, the Depth First Search (DFS) algorithm was successfully implemented and the graph was traversed using depth-first traversal.